BPA-KOM Lab 5 - DHCP

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1. Objective 1

Task Assignment: The aim of this task was to release and renew IP configuration via DHCP and capture communication with the DHCP server in Wireshark.

Solution:

NOTE: for the wireshark portion of this lab, WIFI was used instead of Ethernet, as the lab was performed in an area without ethernet availability. This did not hinder in analyzing the DHCP process.

First, DHCP ip address assignment was confirmed to be true on the pc used for this lab. This is the case, as shown below.

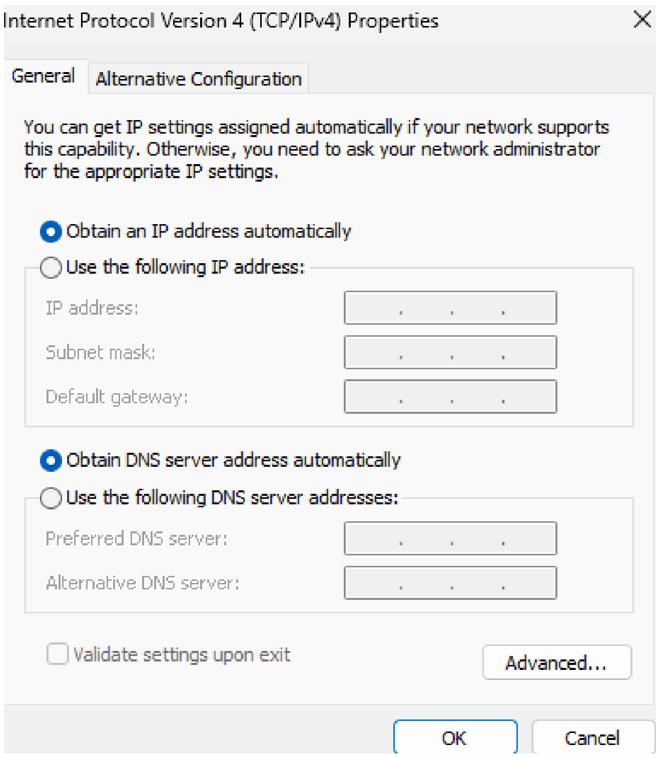

Figure 1: Pc using DHCP IP address assignment.

Then, after packets started being captured in Wireshark, in CMD, the current configuration of the Wifi interface was displayed:

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
   Physical Address. . . . . . . . . : A8-7E-EA-02-3D-ED
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::338:88bf:ab44:4a5a%13(Preferred)
   IPv4 Address. . . . . . . . . . . : 100.69.162.43(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Lease Obtained. . . . . . . . . . : 20 December 2025 13:59:38
   Lease Expires . . . . . . . . . . : 20 December 2025 16:30:08
   Default Gateway . . . . . . . . . : 100.69.160.1
   DHCP Server . . . . . . . . . . . : 100.69.160.1
   DHCPv6 IAID . . . . . . . . . . . : 111705834
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-5D-1A-6B-00-2B-67-B6-37-63
   DNS Servers . . . . . . . . . . . : 147.229.3.200
                                       147.229.3.100
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Figure 2: Current configuration of the Wifi interface.

Then, following the commands `ipconfig /release` and `ipconfig /renew`, the configuration was displayed once again. It should be noted that the IP address assigned remained the same, this is be cause the DHCP server keeps a binding table: it remembers which IP was assigned to which MAC address.

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::338:88bf:ab44:4a5a%13
   IPv4 Address. . . . . . . . . . . : 100.69.162.43
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 100.69.160.1
```

Figure 3: Configuration following appropriate CMD commands.

Then in Wireshark, packets stopped being captured.

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2. Objective 2

Task Assignment: The aim of this task was to analyse DHCP packet content during communication states.

Solution:

After setting the appropriate filter to show captured DHCP packets in wireshark, the following was shown:

```
80170 337.372521   100.69.162.43    100.69.160.1      DHCP    342 DHCP Release  - Transaction ID 0x86ba21dd
80237 345.555310   0.0.0.0          255.255.255.255   DHCP    342 DHCP Discover - Transaction ID 0x175ddcf9
80251 346.202534   100.69.160.1     100.69.162.43     DHCP    320 DHCP Offer    - Transaction ID 0x175ddcf9
80252 346.204283   0.0.0.0          255.255.255.255   DHCP    362 DHCP Request  - Transaction ID 0x175ddcf9
80253 346.207331   100.69.160.1     100.69.162.43     DHCP    320 DHCP ACK      - Transaction ID 0x175ddcf9
```

Figure 4: Captured DHCP packets in Wireshark.

The first packet is the DHCP release message. The Dynamic Host Configuration Protocol section was unrolled, and displayed as follows:

```
Frame 80170: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF_
Ethernet II, Src: Intel_02:3d:ed (a8:7e:ea:02:3d:ed), Dst: Routerboardc_64:db:82 (48:8f:5a:64:db:82)
  Destination: Routerboardc_64:db:82 (48:8f:5a:64:db:82)
  Source: Intel_02:3d:ed (a8:7e:ea:02:3d:ed)
  Type: IPv4 (0x0800)
  [Stream index: 1]
Internet Protocol Version 4, Src: 100.69.162.43, Dst: 100.69.160.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 328
  Identification: 0x369b (13979)
  000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: UDP (17)
  Header Checksum: 0x0000 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 100.69.162.43
  Destination Address: 100.69.160.1
  [Stream index: 62]
User Datagram Protocol, Src Port: 68, Dst Port: 67
  Source Port: 68
  Destination Port: 67
  Length: 308
  Checksum: 0x0bfd [unverified]
  [Checksum Status: Unverified]
  [Stream index: 45]
  [Stream Packet Number: 3]
  [Timestamps]
  UDP payload (300 bytes)
Dynamic Host Configuration Protocol (Release)
  Message type: Boot Request (1)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x86ba21dd
  Seconds elapsed: 0
  Bootp flags: 0x0000 (Unicast)
    0... .... .... .... = Broadcast flag: Unicast
    .000 0000 0000 0000 = Reserved flags: 0x0000
  Client IP address: 0.0.0.0
  Your (client) IP address: 0.0.0.0
  Next server IP address: 0.0.0.0
  Relay agent IP address: 0.0.0.0
  Client MAC address: Intel_02:3d:ed (a8:7e:ea:02:3d:ed)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  Option: (53) DHCP Message Type (Release)
    Length: 1
    DHCP: Release (7)
  Option: (61) Client identifier
    Length: 7
    Hardware type: Ethernet (0x01)
    Client MAC address: Intel_02:3d:ed (a8:7e:ea:02:3d:ed)
  Option: (255) End
    Option End: 255
  Padding: 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
```

Figure 5: DHCP release message.

From the above, the following can be noted:

- Message type: Boot Request
- Client Physical address: a8:7e:ea:02:3d:ed
- Client Logical addresses: 0.0.0.0
- Numeric value of DHCP release message type: 7
- Source port: 68
- Destination port: 67
- Destination IP address: 100.69.160.1
- Destination MAC address: 48:8f:5a:64:db:82

The second packet is the DHCP discover message. The Dynamic Host Configuration Protocol section was unrolled, and displayed as follows:

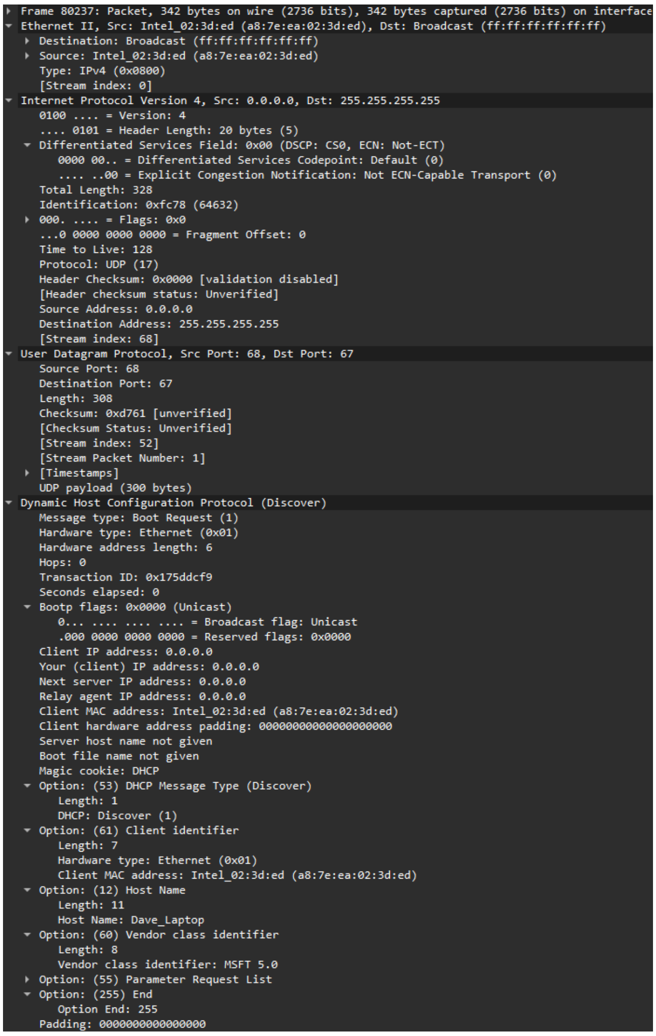
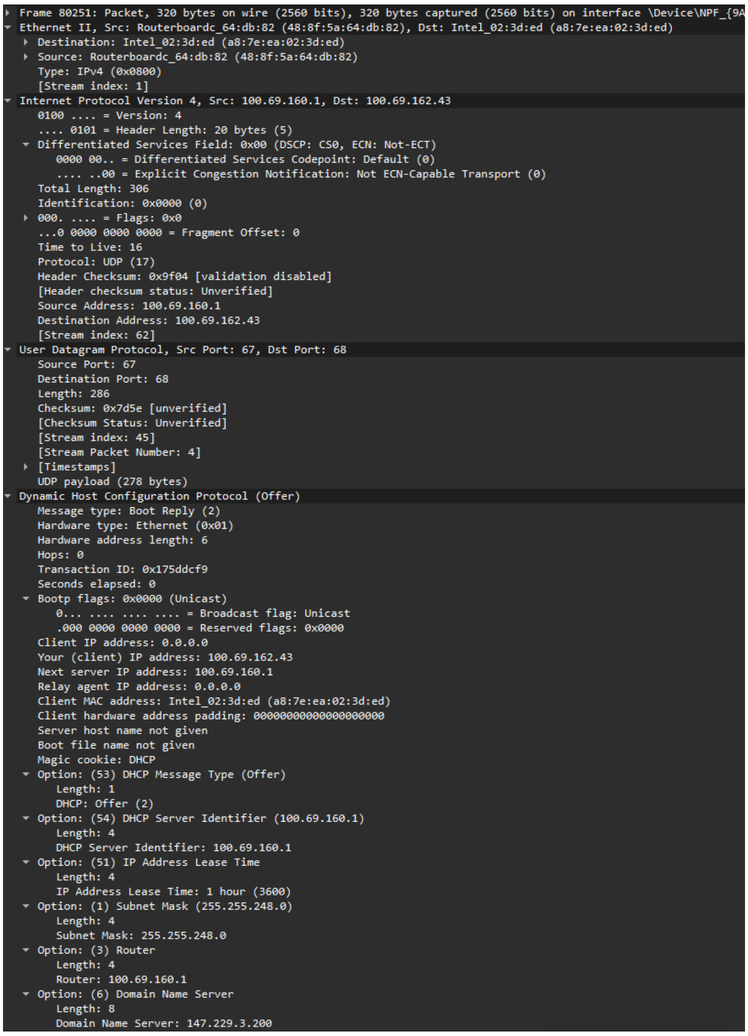


Figure 6: DHCP discover message.

From the above, the following can be noted:

- Message type: Boot Request
- Client Logical addresses: 0.0.0.0
- DHCP server Identifier: No, because client doesn't yet know which server to reference
- Source port: 68
- Destination port: 67
- Destination IP address: 255.255.255.255
- Destination MAC address: ff:ff:ff:ff:ff:ff

The third packet is the DHCP offer message. The Dynamic Host Configuration Protocol section was unrolled, and displayed as follows:



```
Frame 80251: Packet, 320 bytes on wire (2560 bits), 320 bytes captured (2560 bits) on interface \Device\NPF_{9A
Ethernet II, Src: Routerboardc_64:db:82 (48:8f:5a:64:db:82), Dst: Intel_02:3d:ed (a8:7e:ea:02:3d:ed)
  Destination: Intel_02:3d:ed (a8:7e:ea:02:3d:ed)
  Source: Routerboardc_64:db:82 (48:8f:5a:64:db:82)
  Type: IPv4 (0x0800)
  [Stream index: 1]
Internet Protocol Version 4, Src: 100.69.160.1, Dst: 100.69.162.43
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 306
  Identification: 0x0000 (0)
  000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 16
  Protocol: UDP (17)
  Header Checksum: 0x9f04 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 100.69.160.1
  Destination Address: 100.69.162.43
  [Stream index: 62]
User Datagram Protocol, Src Port: 67, Dst Port: 68
  Source Port: 67
  Destination Port: 68
  Length: 286
  Checksum: 0x7d5e [unverified]
  [Checksum Status: Unverified]
  [Stream index: 45]
  [Stream Packet Number: 4]
  [Timestamps]
  UDP payload (278 bytes)
Dynamic Host Configuration Protocol (Offer)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0x175ddcf9
  Seconds elapsed: 0
  Bootp flags: 0x0000 (Unicast)
    0... .... .... .... = Broadcast flag: Unicast
    .000 0000 0000 0000 = Reserved flags: 0x0000
  Client IP address: 0.0.0.0
  Your (client) IP address: 100.69.162.43
  Next server IP address: 100.69.160.1
  Relay agent IP address: 0.0.0.0
  Client MAC address: Intel_02:3d:ed (a8:7e:ea:02:3d:ed)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
  Option: (53) DHCP Message Type (Offer)
    Length: 1
    DHCP: Offer (2)
  Option: (54) DHCP Server Identifier (100.69.160.1)
    Length: 4
    DHCP Server Identifier: 100.69.160.1
  Option: (51) IP Address Lease Time
    Length: 4
    IP Address Lease Time: 1 hour (3600)
  Option: (1) Subnet Mask (255.255.248.0)
    Length: 4
    Subnet Mask: 255.255.248.0
  Option: (3) Router
    Length: 4
    Router: 100.69.160.1
  Option: (6) Domain Name Server
    Length: 8
    Domain Name Server: 147.229.3.200
```

Figure 7: DHCP offer message.

From the above, the following can be noted:

- Message type: Boot Reply
- IP address lease time: 1 hour
- Option Parameters: 53, 54, 51, 1, 3, 6
- Numeric value of DHCP release message type: 7
- Source port: 67
- Destination port: 68
- Destination IP address: 100.69.162.43
- Destination MAC address: a8:7e:ea:02:3d:ed

The fourth packet is the DHCP request message. The Dynamic Host Configuration Protocol section was unrolled, and displayed as follows:

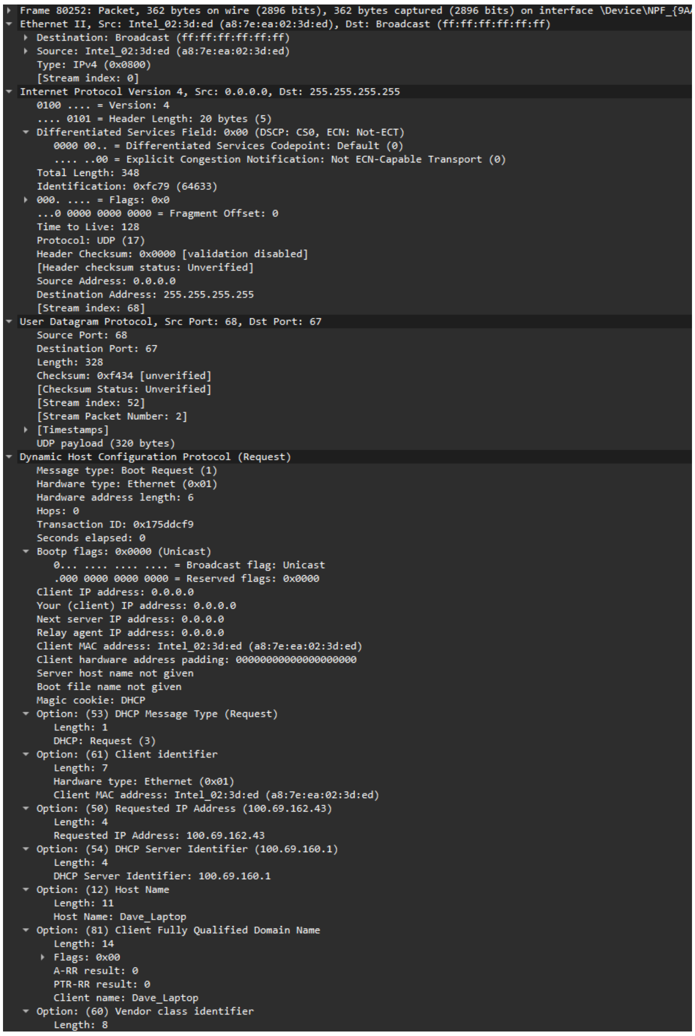


Figure 8: DHCP request message.

From the above, the following can be noted:

- Message type: Boot Request
- IP address specified in options fields: Requested (100.69.162.43), Server Identifier (100.69.160.1)
- Source port: 68
- Destination port: 67
- Destination IP address: 255.255.255.255
- Destination MAC address: ff:ff:ff:ff:ff:ff

The fifth packet is the DHCP ACK message. The Dynamic Host Configuration Protocol section was unrolled, and displayed as follows:

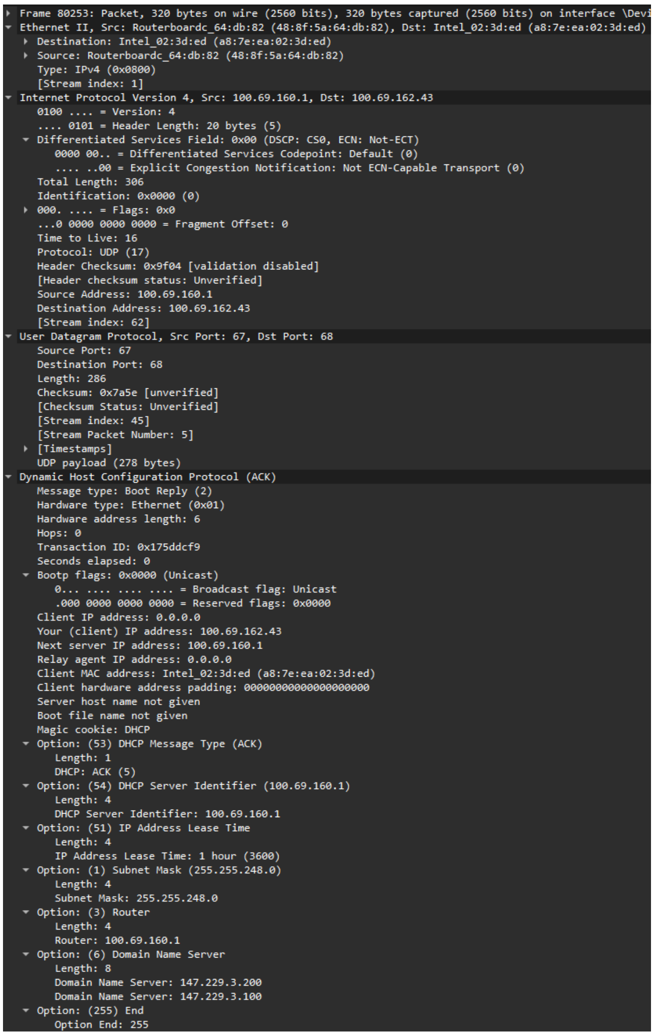


Figure 9: DHCP ACK message.

From the above, the following can be noted:

- Message type: Boot Reply

- Changes with offer message: Destination IP address, source port, destination port.
- Source port: 68
- Destination port: 67
- Destination IP address: 100.69.162.43
- Destination MAC address: a8:7e:ea:02:3d:ed

Throughout the entire communication, it can be noted that the transaction ID was the same for all messages besides for the release message. For the release message, transaction ID was 0x86ba21dd, while for the rest it was 0x175ddcf9.

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3. Objective 3

Task Assignment: The aim of this task was to generate graphs from the captured communication.

Solution:

From the Wireshark graph generation software, the graphs for the packets and bytes sent during the DHCP communication were generated:

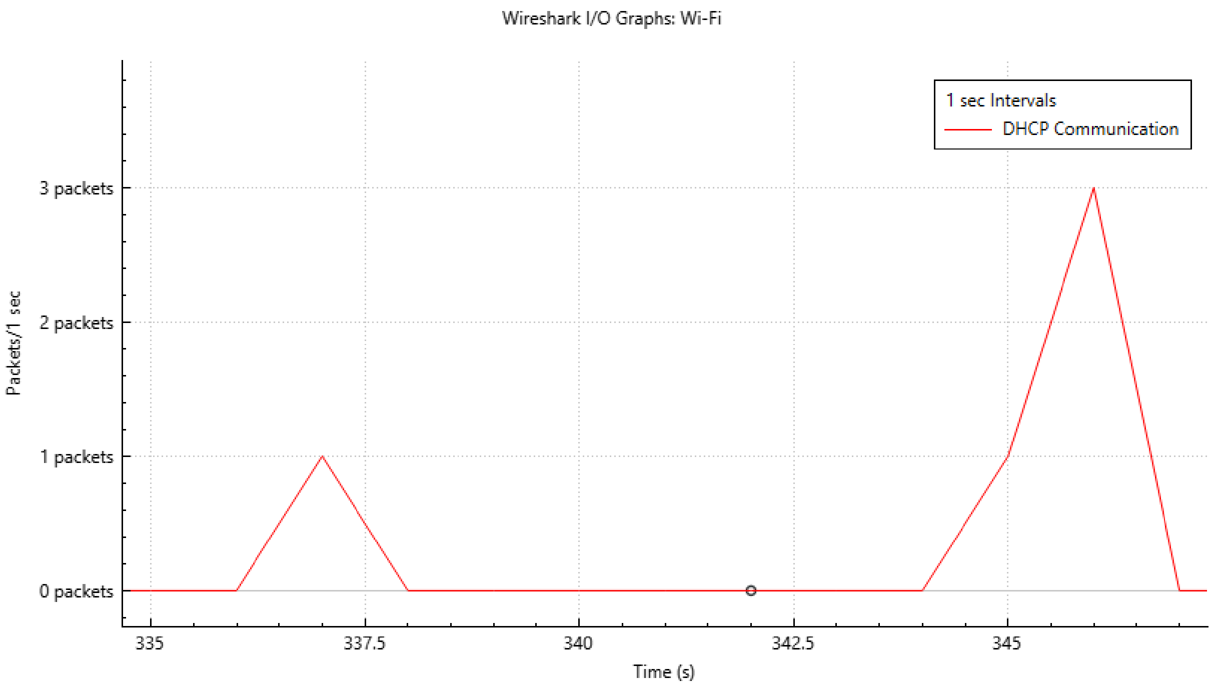

Figure 10: Packets sent during the DHCP communication.

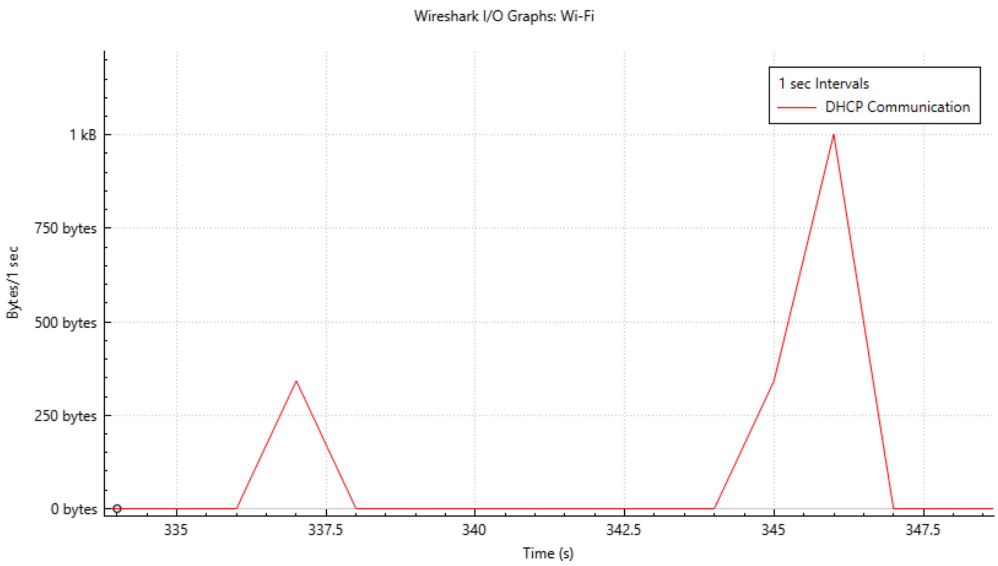

Figure 11: Bytes sent during the DHCP communication.

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4. Objective 4

Task Assignment: The aim of this task was to create the topology in Cisco Packet Tracer.

Solution:

After adding the necessary components specified in the lab, the topology for the rest of the lab is as follows:

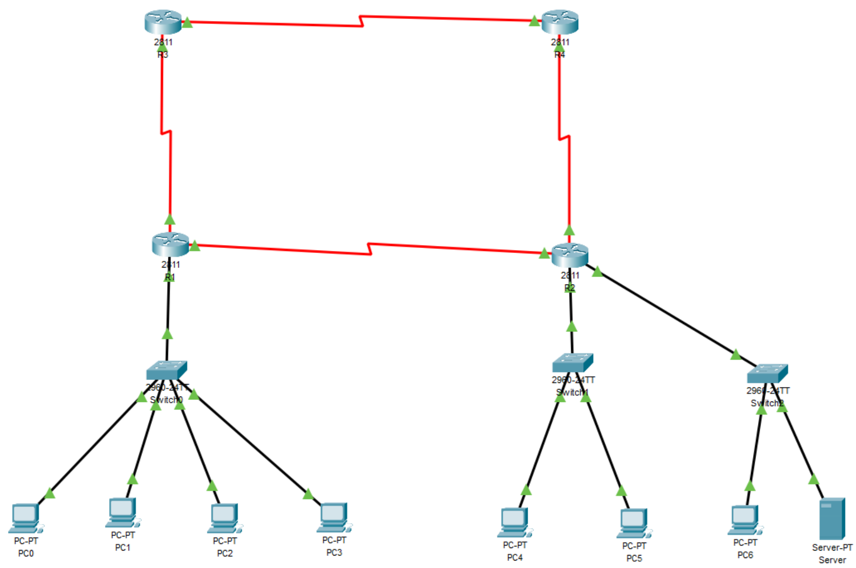

Figure 12: Topology in Cisco Packet Tracer.

After setting the topology, connectivity was verified from the PCs:

```
C:\> ping 172.20.3.1

Pinging 172.20.3.1 with 32 bytes of data:

Reply from 172.20.3.1: bytes=32 time<1ms TTL=128
Reply from 172.20.3.1: bytes=32 time<1ms TTL=128
Reply from 172.20.3.1: bytes=32 time=1ms TTL=128
Reply from 172.20.3.1: bytes=32 time<1ms TTL=128

Ping statistics for 172.20.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 13: Verifying connectivity with ping.

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5. Objective 5

Task Assignment: The aim of this task was to configure R2 as the DHCP server for both connected LANs. Additionally, PCs were set to use DHCP and analyse communication.

Solution:

Setting R2 as the DHCP server for both connected LANs involved defining two pools on the router, one for the LAN with `172.20.1/24` address space and one for `172.20.3.0/24` address space.

First, the addresses which shouldn't be assigned by the DHCP server are explicitly defined. In total, 12 total addresses are excluded by using the following commands:

- `R2(config)#ip dhcp excluded-address 172.20.1.254`
- `R2(config)#ip dhcp excluded-address 172.20.3.1 172.20.3.10`
- `R2(config)#ip dhcp excluded-address 172.20.3.254`

Then, the individual pools are defined by issuing the `ip dhcp pool` command, and assigning the appropriate network IP address and subnet mask to determine the address space for the given pool. Additionally, a default-router and dns-server are assigned.

After the above is sorted, in simulation mode, one PC from the client network has their IP configuration changed from static to dynamic. This generates the DHCP discover message, as follows:

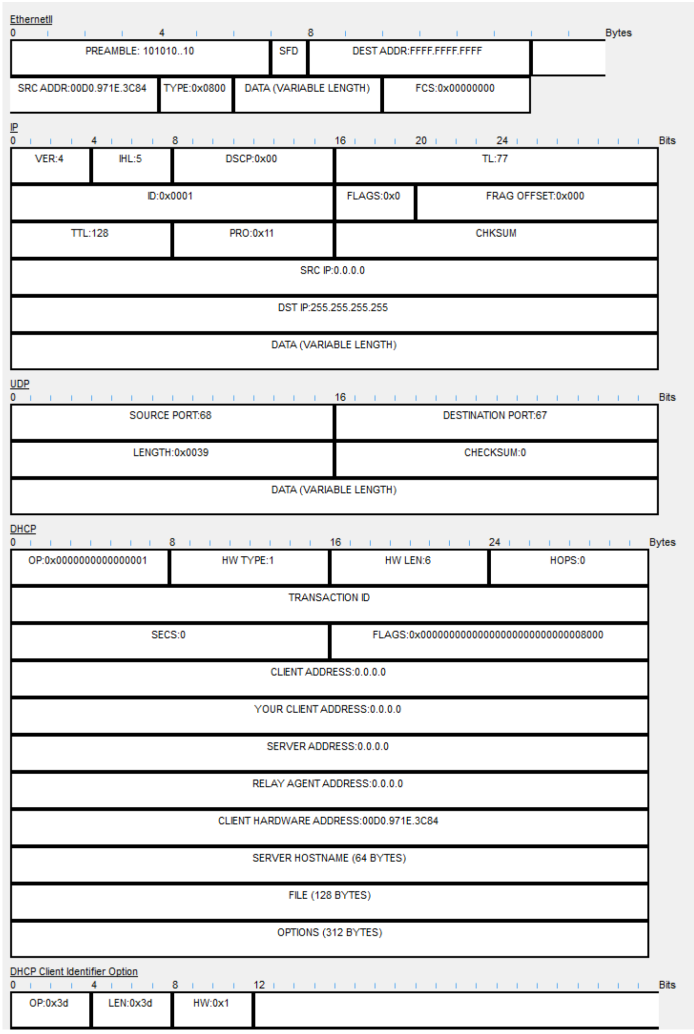

Figure 14: DHCP discover message.

From the above, the following can be noted:

- IP addresses set in DHCP section: Client address, server address, relay agent address
- Source port: 68
- Destination port: 67
- Destination IP address: 255.255.255.255
- Destination MAC address: FFFF.FFFF.FFFF

Then, Capture then forward is clicked until the DHCP offer message is generated.

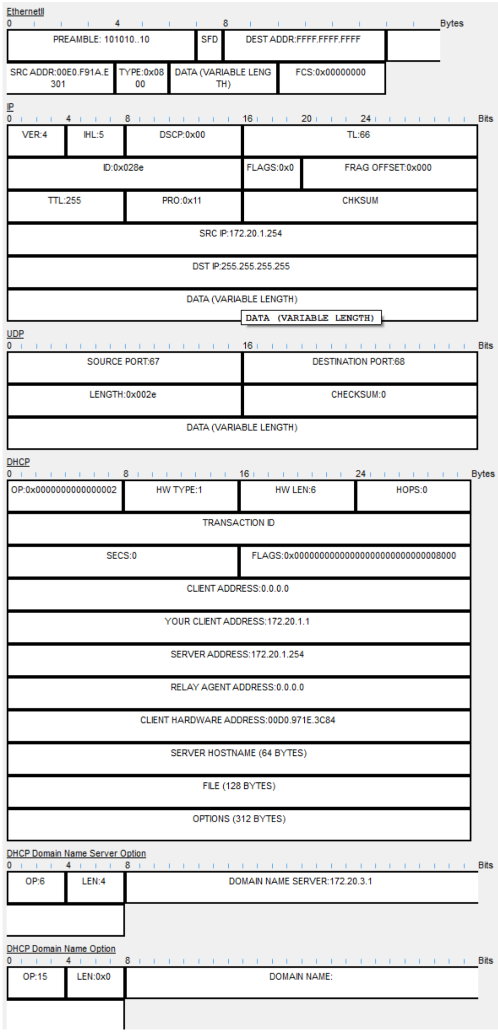

Figure 15: DHCP offer message.

From the above, the following can be noted:

- Pool used to assign the IP configuration: CLIENT-NETWORK
- IP addresses router provides to the client: 172.20.1.1, 0.0.0.0
- Source port: 67
- Destination port: 68
- Destination IP address: 255.255.255.255
- Destination MAC address: FFFF.FFFF.FFFF

Then, Capture then forward is clicked until the DHCP Request message is generated.

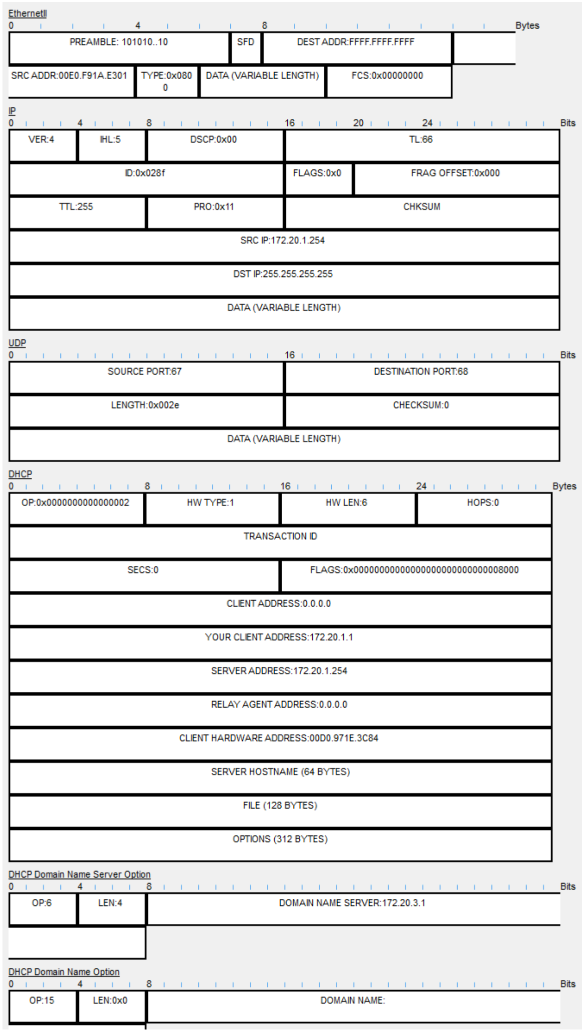

Figure 16: DHCP Request message.

From the above, the following can be noted:

- IP addresses specified by the client: 172.20.1.1, 0.0.0.0
- Destination IP address: 255.255.255.255
- Destination MAC address: FFFF.FFFF.FFFF

Then, Capture then forward is clicked until the DHCP ACK message is generated.

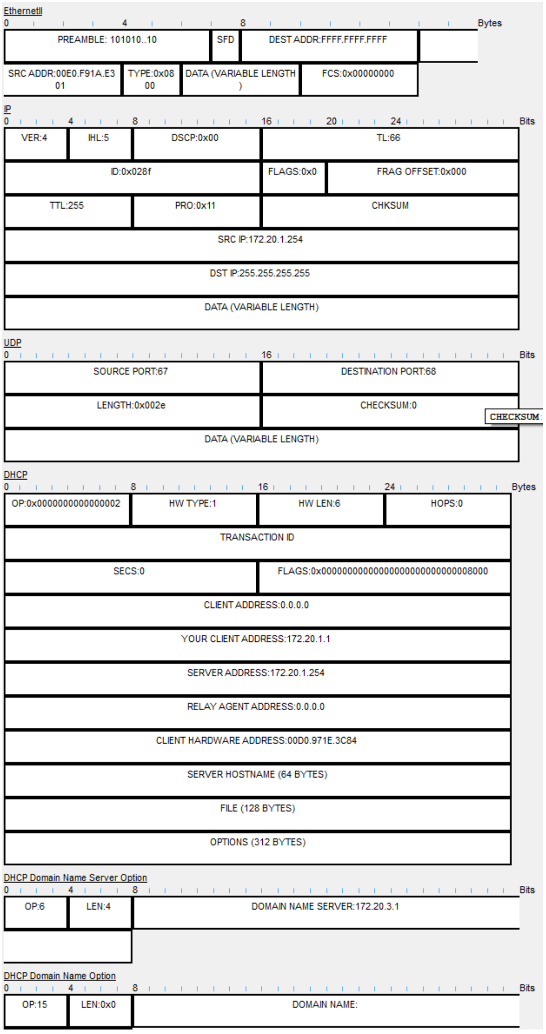

Figure 17: DHCP ACK message.

From the above, the following can be noted:

- How packet content differs from DHCP offer: It does not differ
- Destination IP address: 255.255.255.255
- Destination MAC address: FFFF.FFFF.FFFF

Capture then forward is clicked until the ACK message reaches the client. This applies the configuration as shown below:

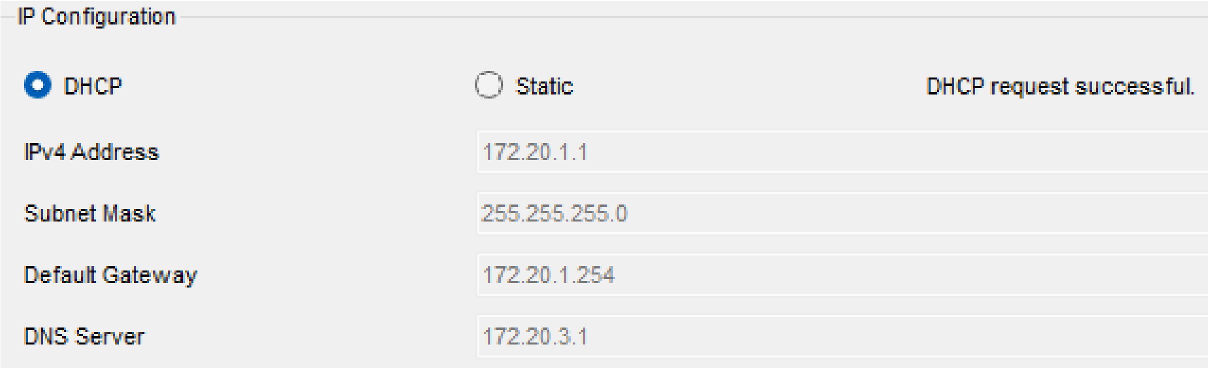

Figure 18: Configuration of DHCP client.

To verify the correct configuration, in realtime mode, the server's web page *www.mywebpage.com* is accessed.

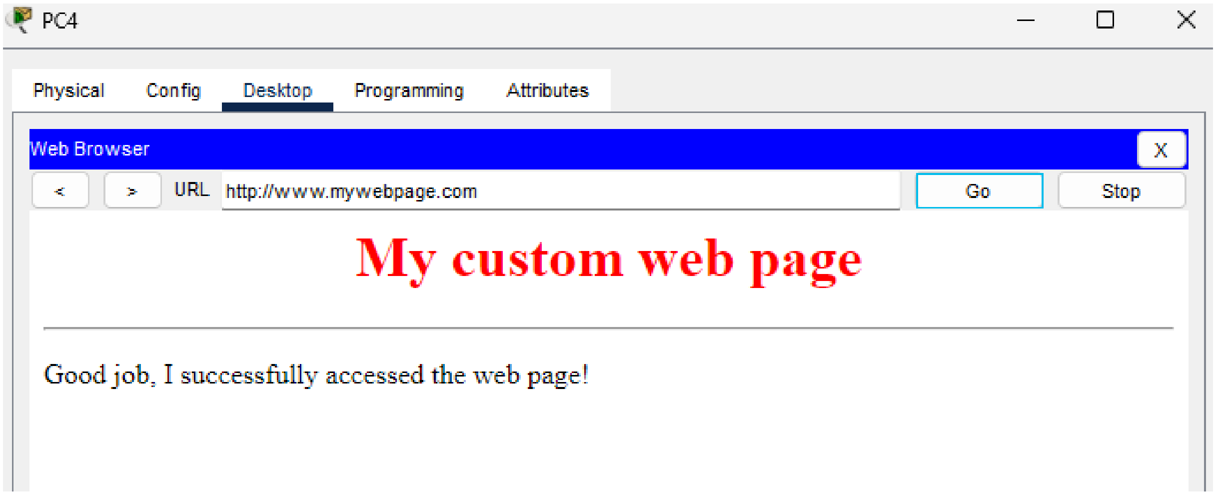

Figure 19: *www.mywebpage.com*.

The above process is repeated for the other PC in the same subnet. The IP address it is assigned can be seen below.

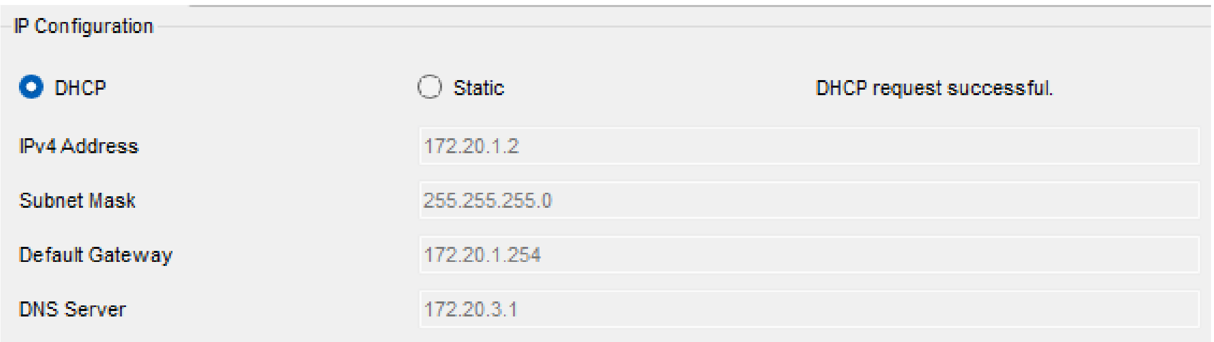

Figure 20: Result of DCHP communication on other PC in same subnet.

The above process is repeated for the administrator's PC in the server network. The IP address it is assigned can be seen below. The pool used to choose the address is SERVER-NETWORK.

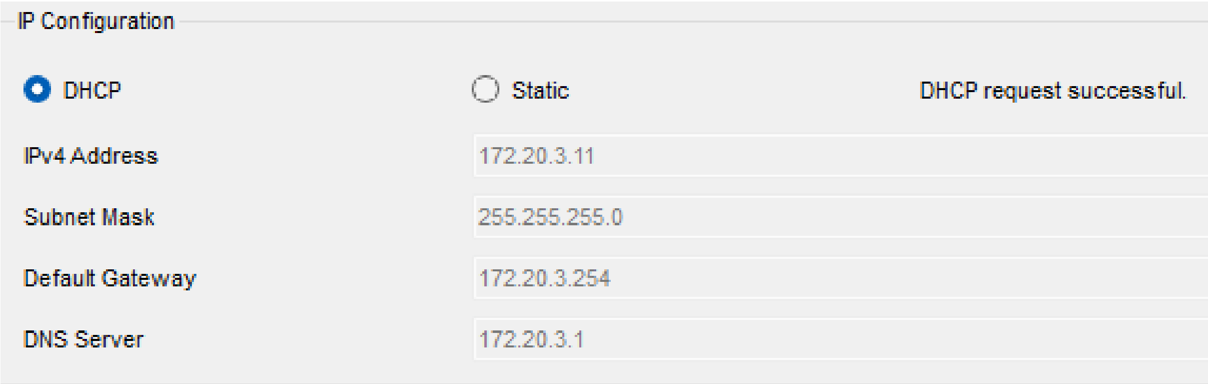

Figure 21: Result of DCHP communication on the Administrator's PC.

Finally, in realtime mode, connectivity is verified between the devices as shown below:

```
C:\>ping 172.20.3.11

Pinging 172.20.3.11 with 32 bytes of data:

Request timed out.
Reply from 172.20.3.11: bytes=32 time<1ms TTL=127
Reply from 172.20.3.11: bytes=32 time=1ms TTL=127
Reply from 172.20.3.11: bytes=32 time<1ms TTL=127

Ping statistics for 172.20.3.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss)
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.20.3.11

Pinging 172.20.3.11 with 32 bytes of data:

Reply from 172.20.3.11: bytes=32 time<1ms TTL=127
Reply from 172.20.3.11: bytes=32 time<1ms TTL=127
Reply from 172.20.3.11: bytes=32 time<1ms TTL=127
Reply from 172.20.3.11: bytes=32 time<1ms TTL=127

Ping statistics for 172.20.3.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 22: Verification of connectivity using ping.

6. Objective 6

Task Assignment: The aim of this task was to enable DHCP service on the LAN server and configure R2 as the DHCP relay agent. Additionally, the IP configuration was released and renewed via DHCP and the role of the relay agent was explored.

Solution:

Before anything else is done, DHCP service on the R2 router is disabled using the `no service dhcp` command in Configuration mode. This preserves the service while turning it off. This change was verified by switching the administrator's PC configuration to Static and back to dynamic. The result is the following:

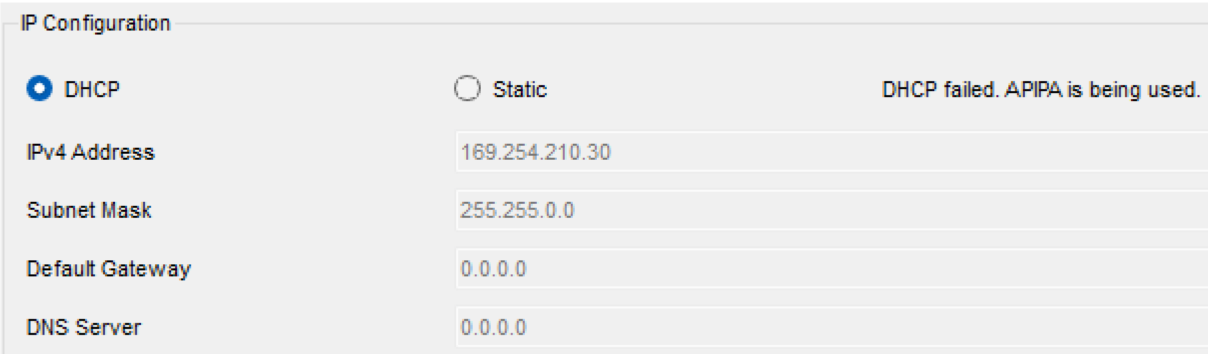

Figure 23: New IP address assignment to administrator PC.

After, the service is configured on the Server device. Pools are created for both the server network and the client network.

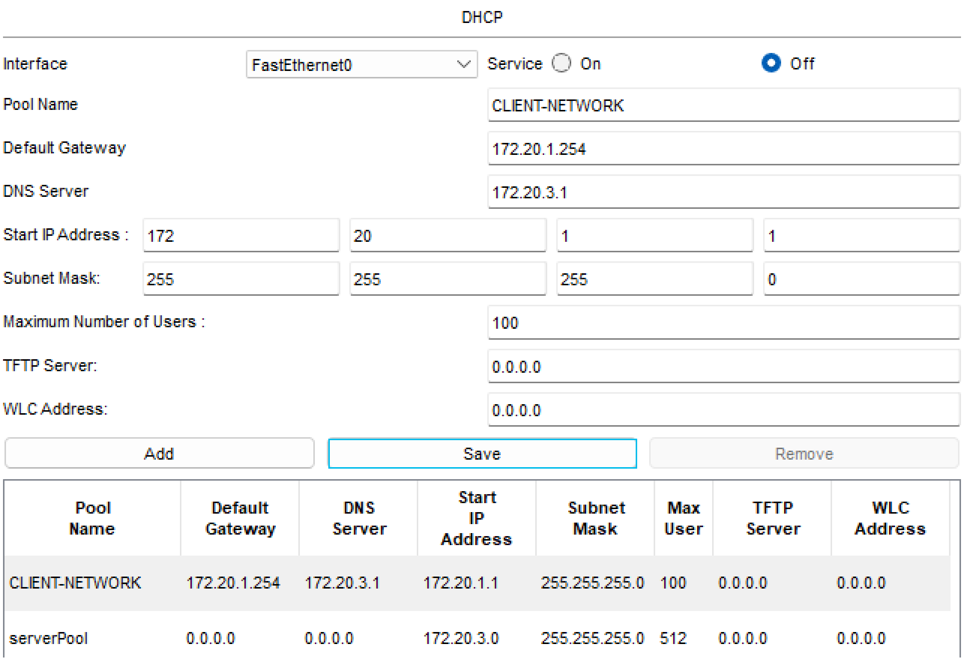

Figure 24: Client-Network Pool setup.

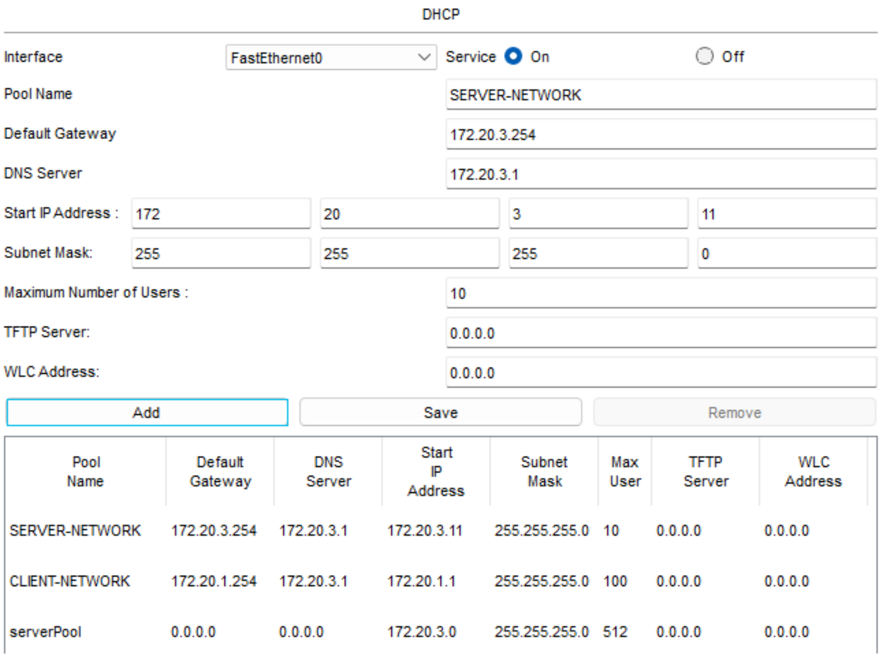

Figure 25: Server-Network Pool setup.

Then, on the administrator's PC, the IP configuration is set to Static and then back to dynamic. It then receives the first host address specified in the pool, as shown below:

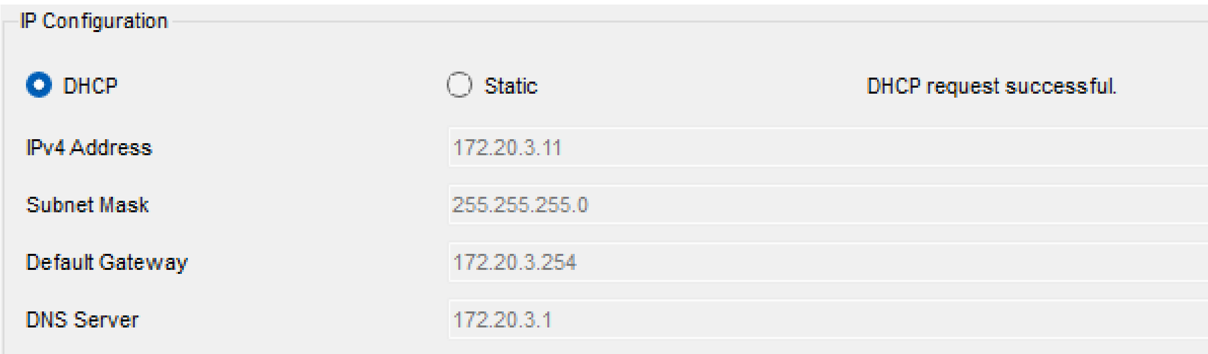

Figure 26: Administrator PC receiving first host address specified in the pool.

DHCP configuration is reloaded on the PCs in the client network in the same way.

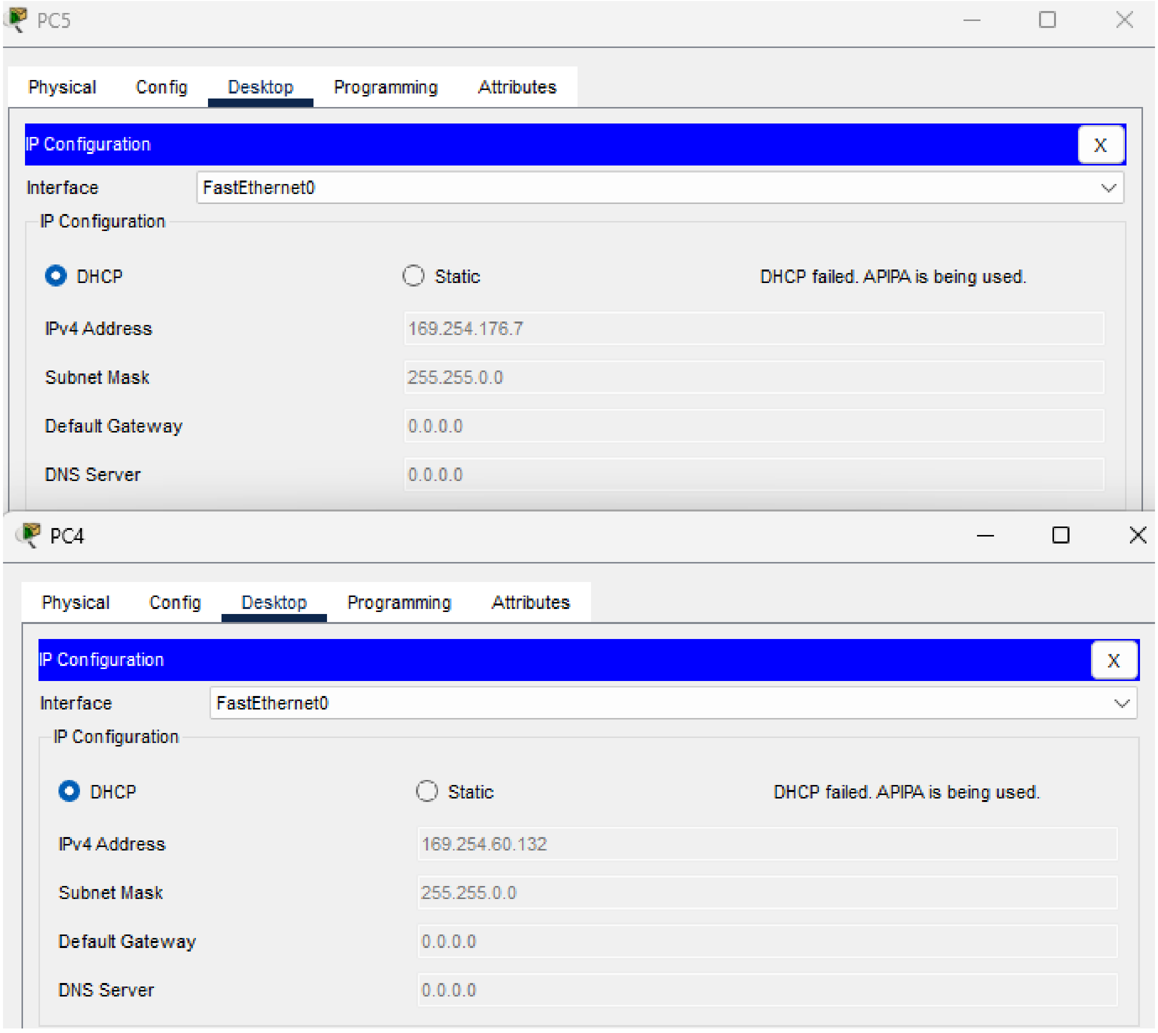

Figure 27: DHCP configuration reloaded on the PCs in the client network.

The PCs received addresses from the APIPA range, this is because currently the only DHCP server is in the different network. It is for this reason that the router is now to be configured to the role of the DHCP relay agent.

After this is done using the commands below, the router is now capable of forwarding DHCP messages arriving at the FastEthernet 0/0 interface to the DHCP server with 172.20.3.1 address.

- `R2(config)#interface FastEthernet0/0`
- `R2(config-if)#ip helper-address 172.20.3.1`

After switching to simulation mode, and enabling one PC from the client-network to obtain the DHCP configuration, the DHCP discover message is generated. Capture then forward is clicked until the packet reaches the router.

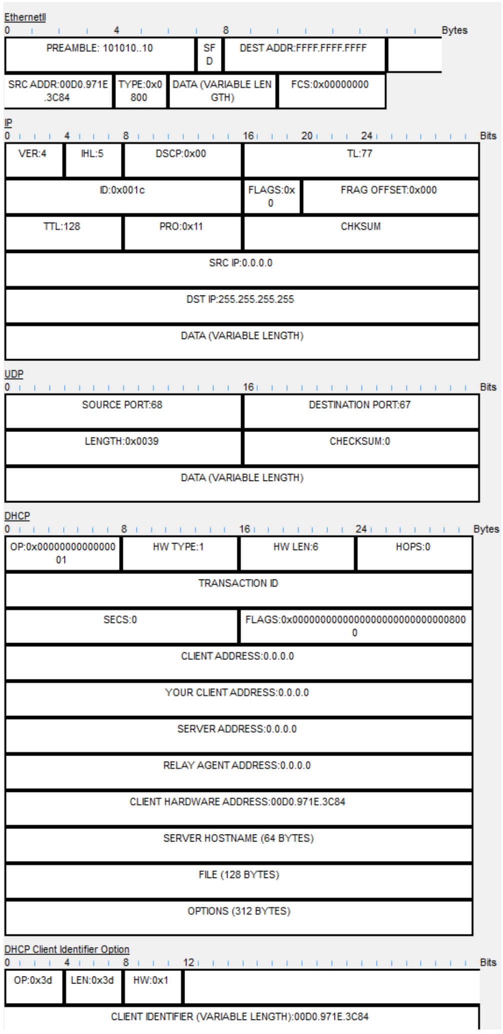

Figure 28: DHCP discover message at router: Inbound PDU details.

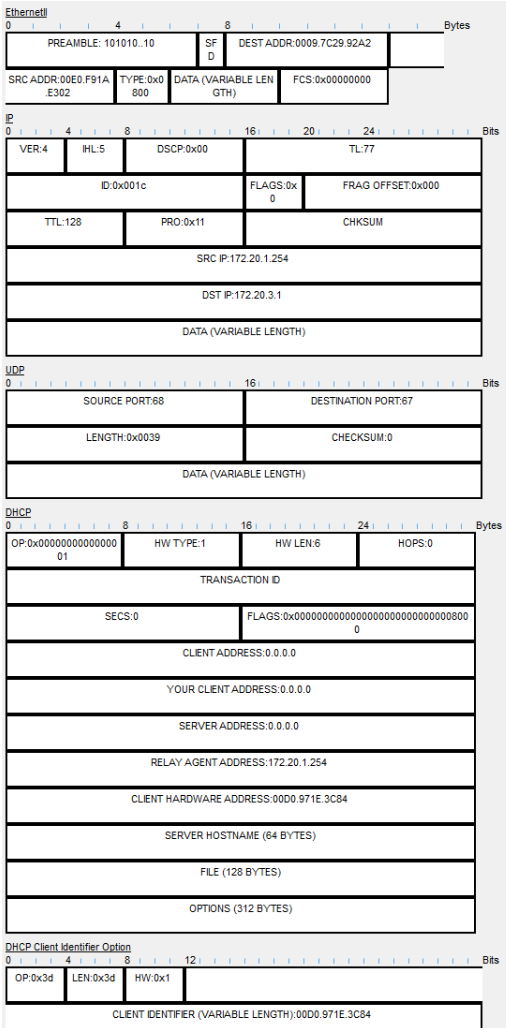

Figure 29: DHCP discover message at router: Outbound PDU details.

From the above, the following can be noted:

- Changes in the DHCP section: relay agent address in outbound is not 0.0.0.0
- Source MAC address (inbound): 00D0.971E.3C84
- Source MAC address (outbound): 00E0.F91A.E302
- Destination MAC address (inbound): FFFF.FFFF.FFF
- Destination MAC address (outbound): 0009.7C29.92A2
- Source IP address (inbound): 0.0.0.0
- Source IP address (outbound): 172.20.1.254
- Destination IP address (inbound): 255.255.255.255
- Destination IP address (outbound): 172.20.3.1

Then, Capture then forward is clicked until the DHCP discover message reaches the server

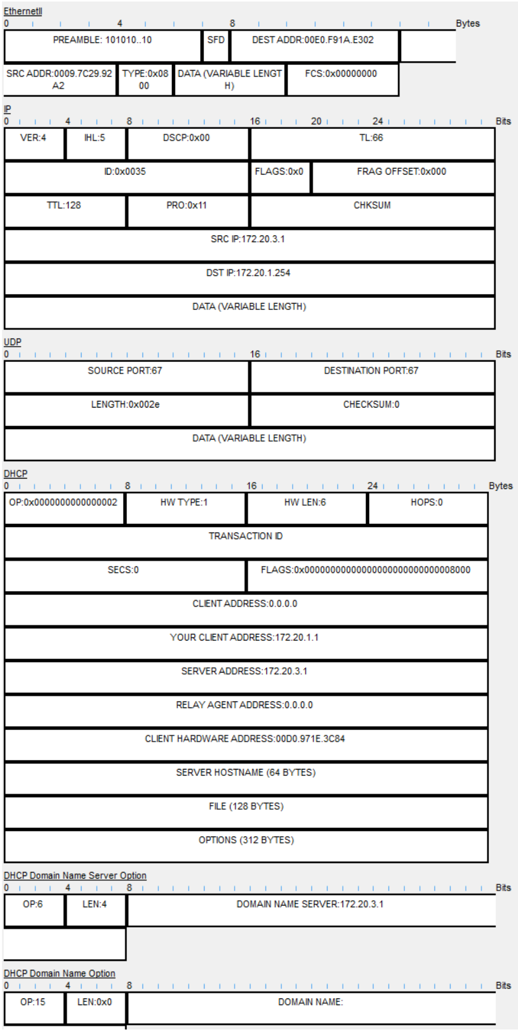

Figure 30: DHCP discover message at server.

From the above, the following can be noted:

- IP addresses included in DHCP section: your client (172.20.1.1), server address (172.20.3.1)
- Destination IP address: 172.20.1.254
- Destination MAC address: 00E0.F91A.E302

Then, Capture then forward is clicked until the DHCP offer message reaches the router.

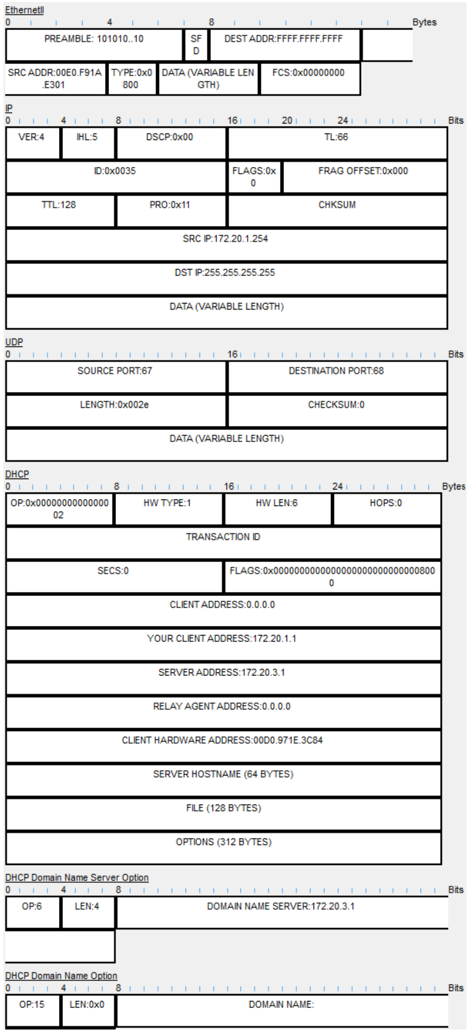

Figure 31: DHCP offer message at router.

From the above, the following can be noted:

- Changes in the DHCP section: no changes between inbound and outbound
- Source MAC address (inbound): 0009.7C29.92A2
- Source MAC address (outbound): 00E0.F91A.E302
- Destination MAC address (inbound): 00E0.F91A.E302
- Destination MAC address (outbound): FFFF.FFFF.FFFF
- Source IP address (inbound): 172.20.3.1
- Source IP address (outbound): 172.20.1.254
- Destination IP address (inbound): 172.20.1.254
- Destination IP address (outbound): 255.255.255.255

Then, Capture then forward is clicked until the DHCP offer message reaches the PC and a request message is generated.

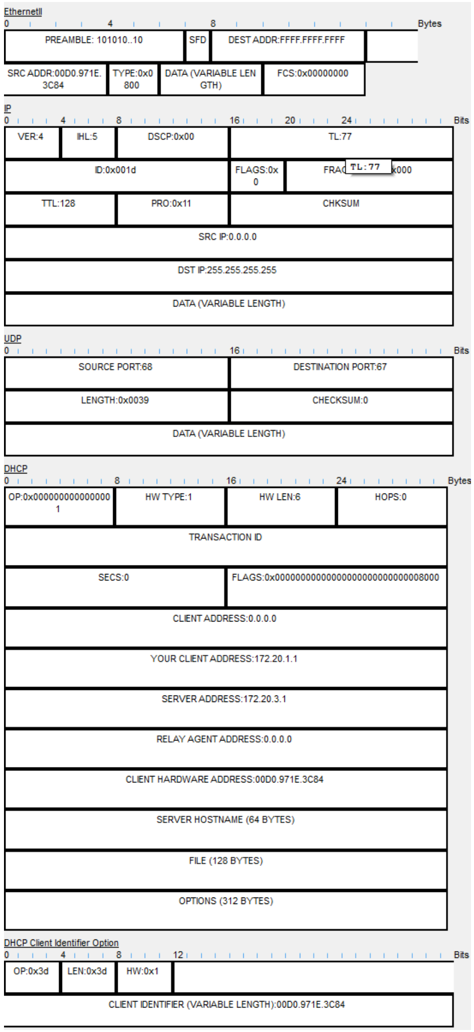

Figure 32: DHCP request message generated at PC.

From the above, the following can be noted:

- How packet content differs from the inbound packet: OP is different (last number 1 instead of 2), and no DHCP domain name area
- Destination IP address: 255.255.255.255
- Destination MAC address: FFFF.FFFF.FFFF

Then, Capture then forward is clicked until the packet reaches the router once again.

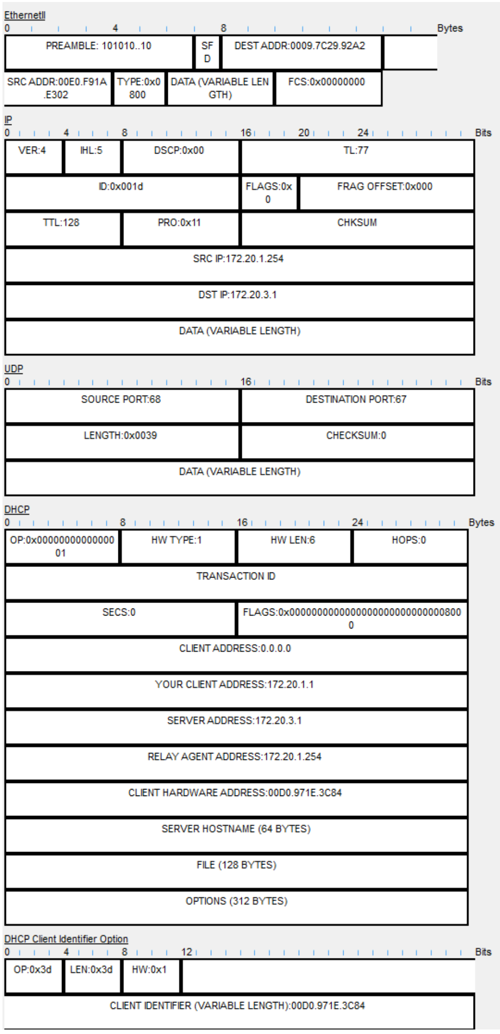

Figure 33: DHCP message at the router.

From the above, the following can be noted:

- How packet content differs from DHCP inbound packeter: The relay agent address is now filled instead of being 0.0.0.0
- Destination IP address: 172.20.3.1
- Destination MAC address: 0009.7C29.92A2

Then, Capture then forward is clicked until the DHCP message reaches the server and the Ack message is generated.

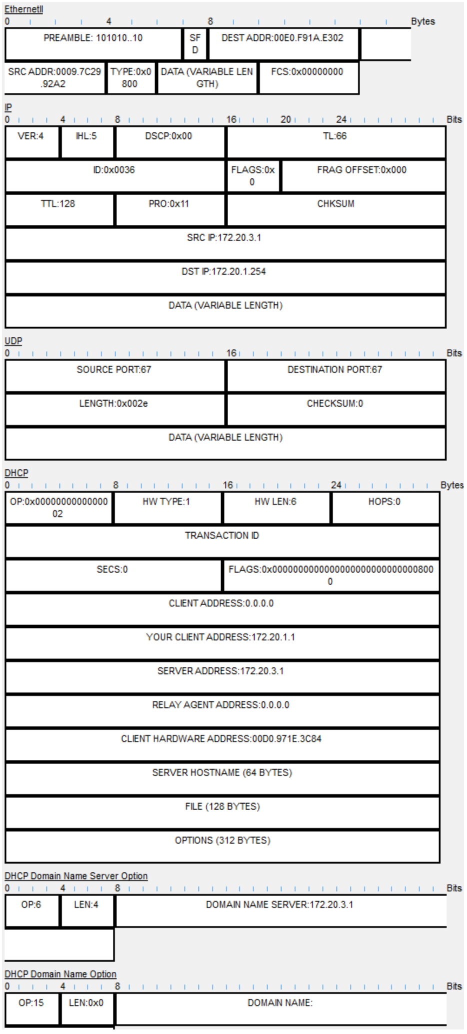

Figure 34: DHCP ACK message at the server.

From the above, the following can be noted:

- How packet content differs from DHCP offer: Although some vaues are different, there are no additional parameters in comparison with the offer
- Destination IP address: 172.20.1.254
- Destination MAC address: 00E0.F91A.E302

Capture then forward is clicked until the DHCP message reaches the router

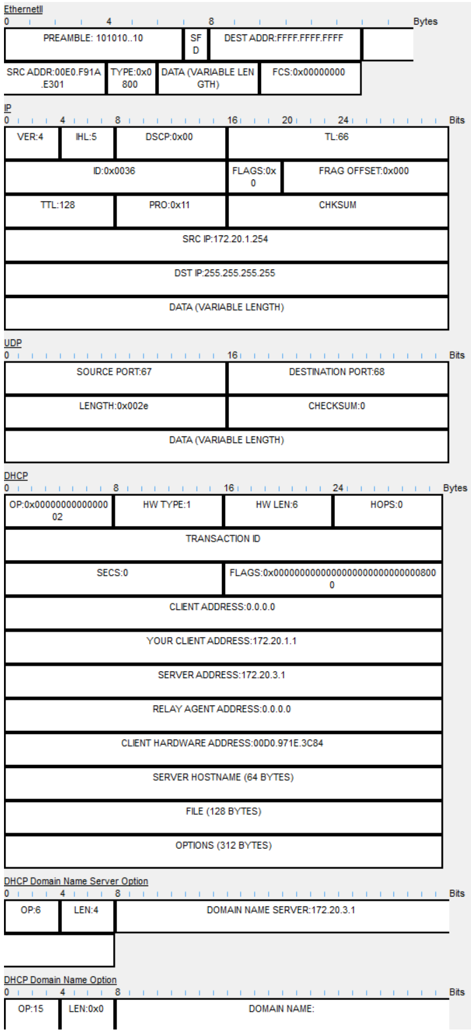

Figure 35: DHCP message at the router.

- Source MAC address: 00E0.F91A.E301
- Destination MAC address: FFFF.FFFF.FFFF
- Source IP address: 172.20.1.254
- Destination IP address: 255.255.255.255

Finally, capture then forward is clicked until the packet reaches the PC. Realtime mode is enabled, and connectivity is verified by accessing `www.mywebpage.com`. The configuration set and the accessing of the webpage are shown below.

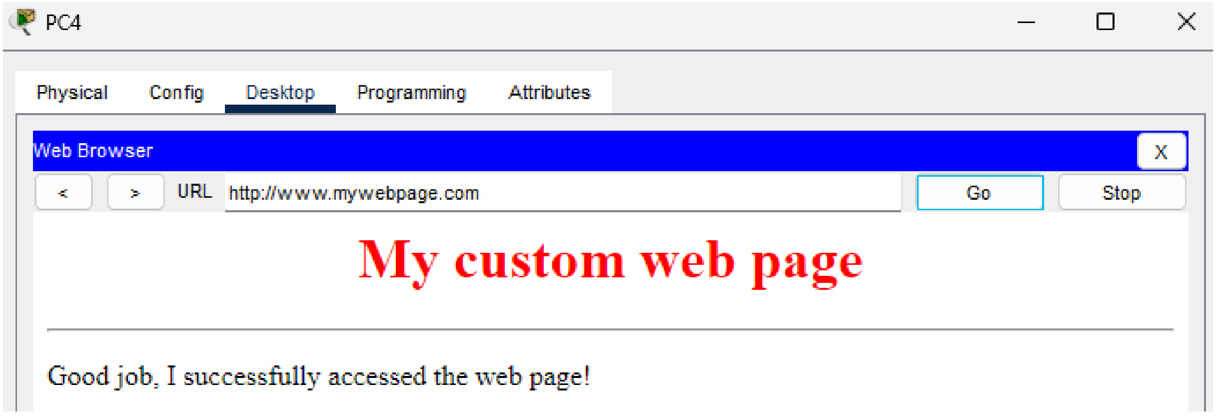

Figure 36: Website successfully accessed.

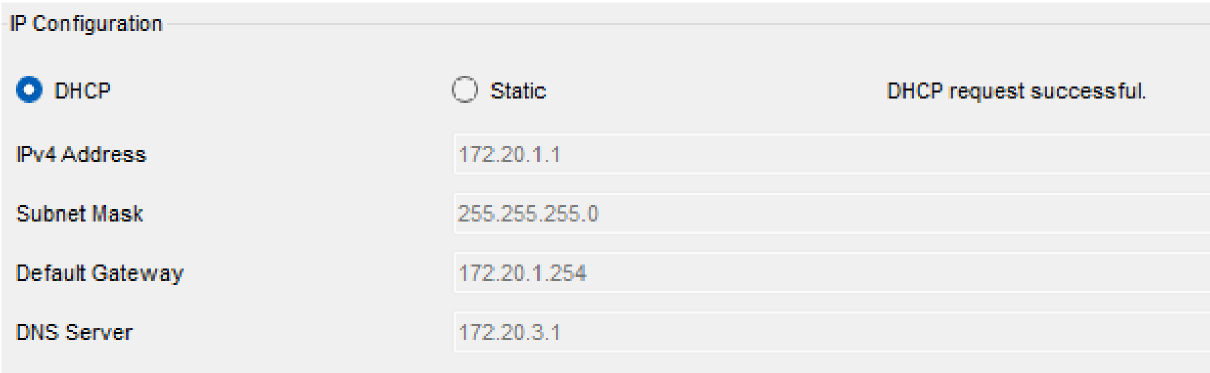

Figure 37: Current configuration of the PC.

The IP address of the other PC in the same subnet is as shown below.

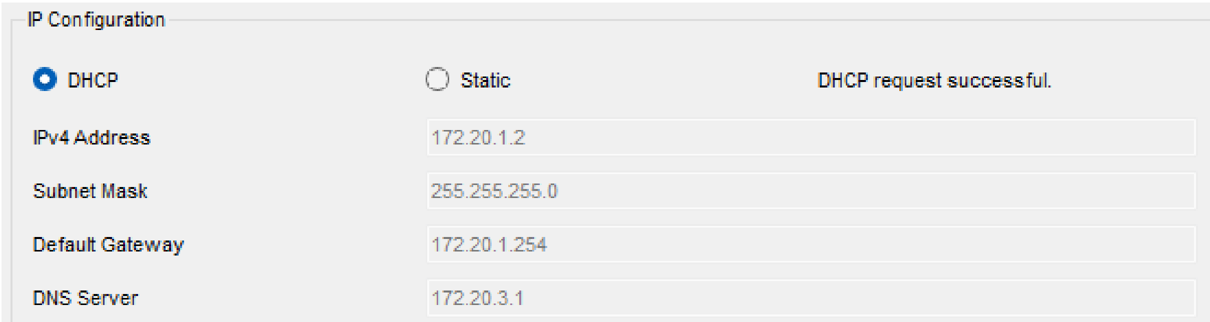

Figure 38: IP address of the other PC in the same subnet.

Connectivity is finally verified between other devices via ping

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.20.3.11

Pinging 172.20.3.11 with 32 bytes of data:

Reply from 172.20.3.11: bytes=32 time<1ms TTL=127
Reply from 172.20.3.11: bytes=32 time<1ms TTL=127
Reply from 172.20.3.11: bytes=32 time<1ms TTL=127
Reply from 172.20.3.11: bytes=32 time<1ms TTL=127

Ping statistics for 172.20.3.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 39: Successful connectivity verification via ping.

---

7. Final Questions

Question 1: How many messages are exchanged during the initial address assignment? What are they called?
The initial DHCP address assignment consists of **four** messages.

- **DHCP Discover:** sent by the client to locate available DHCP servers
- **DHCP Offer:** sent by the server offering an IP address and configuration
- **DHCP Request:** sent by the client to request the offered IP address
- **DHCP Acknowledgment (ACK):** sent by the server to confirm the address assignment

Question 2: What is defined by the lease time?
The lease time defines:

- The duration for which an IP address is valid and assigned to a client
- The time after which the client must renew the lease or release the address back to the DHCP server

Question 3: What is the relay agent used for?
A DHCP relay agent is used to forward DHCP messages between clients and servers located on different networks

Question 4: When a relay agent is used in the DHCP communication, how does a server identify which pool to use to assign the address?
The DHCP server identifies the correct address pool by:

- Examining the Gateway IP address field
- Using the Gateway IP address value to determine the subnet of the requesting client and selecting the appropriate DHCP pool

Question 5: How does a relay agent alter destination IP and MAC addresses while forwarding the DHCP messages?
When forwarding DHCP messages, the relay agent:

- Changes the **destination IP address** from the broadcast address to the DHCP server's unicast IP address
- Updates the **destination MAC address** to the MAC address of the next-hop device toward the server
- Inserts its own IP address into the Gateway IP address field before forwarding the message

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